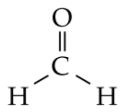
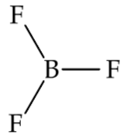
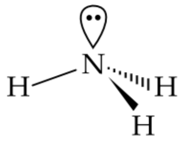
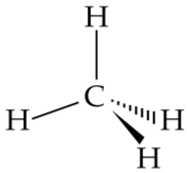
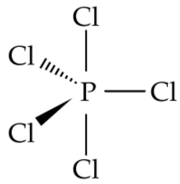
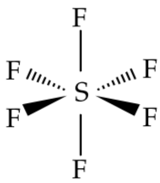
Intensive note (Topic 6: Microscopic world II)

Molecular shape

Bond pair	Lone pair	Molecular shape	Example	Atomicity
2	0	Linear	O＝C＝O (Obeying octet rule) F－Be－F *(NOT obeying octet rule)	3
2	2	V-shaped	 (Obeying octet rule)	3
3	0	Trigonal planar	 (Obeying octet rule)  *(NOT obeying octet rule)	4
3	1	Trigonal pyramidal	 (Obeying octet rule)	4
4	0	Tetrahedral	 (Obeying octet rule)	5
5	0	Trigonal bipyramidal	 (NOT obeying octet rule)	6
6	0	Octahedral	 (NOT obeying octet rule)	7

*For non-octet molecules:

- Period 2 atoms can only have less than 8 outermost shell electrons (OF_6, NF_5 do NOT exist)
- Period 3 or above atoms can have more than or less than 8 outermost shell electrons. (Usually more than 8)

Example

In a H_2O molecule, there are **2 bond pairs** of electrons and **2 lone pairs** of electrons around the **central atom O**.

To **minimize the repulsion between electron pairs**, a H_2O molecule is **V-shaped**

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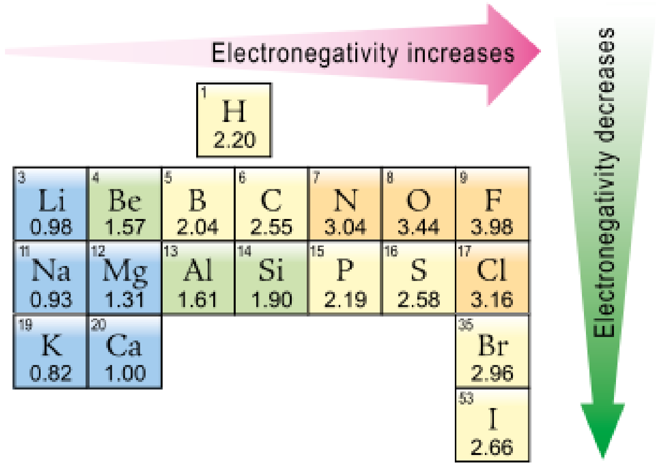
Intensive note (Topic 6: Microscopic world II)

Electronegativity

Electronegativity is a power of the atom to attract electrons to itself in a bonding situation

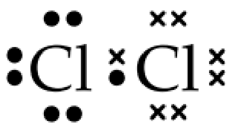


- Across a period, the electronegativity of the main group elements increases from left to right.
- Down a group, the electronegativity of the main group elements generally decreases.

Polar bond

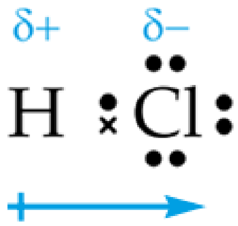
- A covalent bond is polar when the bonding electrons are not equally shared between the bonded atoms. This results when two bonded atoms have different electronegativities.

Non-polar bond



As both Cl atoms have the same electronegativity, Cl–Cl bond is a non-polar bond

Polar bond



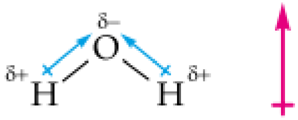
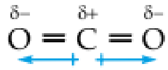
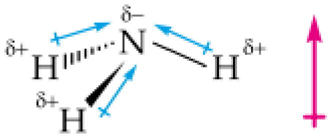
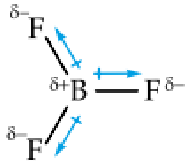
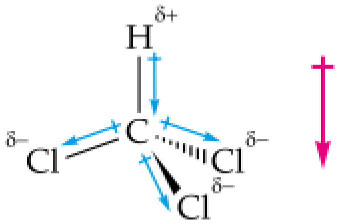
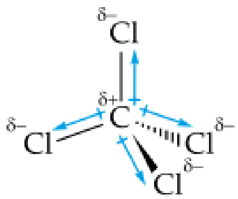
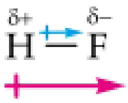
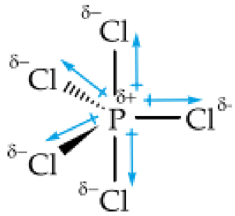
As H and Cl have different electronegativities, H–Cl bond is a polar bond. Since Cl is more electronegative, the shared electrons are attracted more strongly towards Cl.

Note: If the electronegativity difference between 2 atoms are very large, ionic bond will be formed (Transfer of electron)

Polar molecule

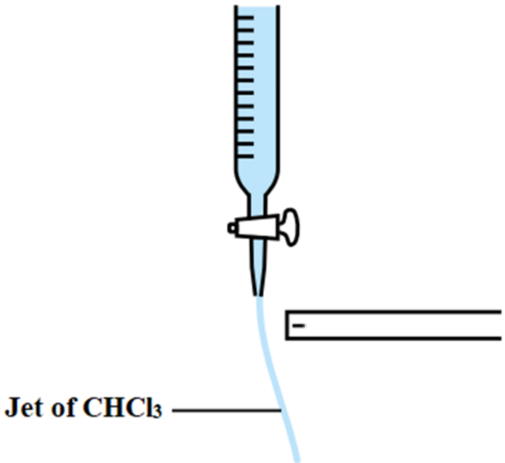
Criteria of being a non-polar molecule (DSE syllabus only)	
A. Molecules without polar bonds e.g. H_2, N_2, O_2, F_2, Cl_2, Br_2, I_2 P_4, S_8, Ne, Ar, He	B. Molecules with polar bonds but fulfilling the following: 1. No lone pair of electrons on central atom 2. All polar bonds are the same e.g. CO_2, BF_3, CH_4, PCl_5, SF_6

Intensive note (Topic 6: Microscopic world II)

Polar molecules	Non-polar molecules
	
Molecular shape of H_2O is V-shape Polar O-H bonds are not arranged symmetrically **Polarities of polar O-H bonds cannot cancel out.**	Molecular shape of CO_2 is linear Polar C=O bonds are arranged symmetrically **Polarities of polar C=O bonds can cancel out**
	
Molecular shape of NH_3 is trigonal pyramidal Polar N-H bonds are not arranged symmetrically **Polarities of polar N-H bonds cannot cancel out.**	Molecular shape of BF_3 is trigonal planar Polar B-F bonds are arranged symmetrically **Polarities of polar B-F bonds can cancel out**
	
Molecular shape of $CHCl_3$ is tetrahedral Polarities of polar C-H and C-Cl bonds are different **Polarities of polar C-H and C-Cl bonds cannot cancel out**	Molecular shape of CCl_4 is tetrahedral Polar C-Cl bonds are arranged symmetrically **Polarities of polar C-Cl bonds can cancel out**
	
Each HF molecule has only one polar H–F bond.	Molecular shape of PCl_5 is trigonal bipyramidal Polar P-Cl bonds are arranged symmetrically **Polarities of polar P-Cl bonds can cancel out**

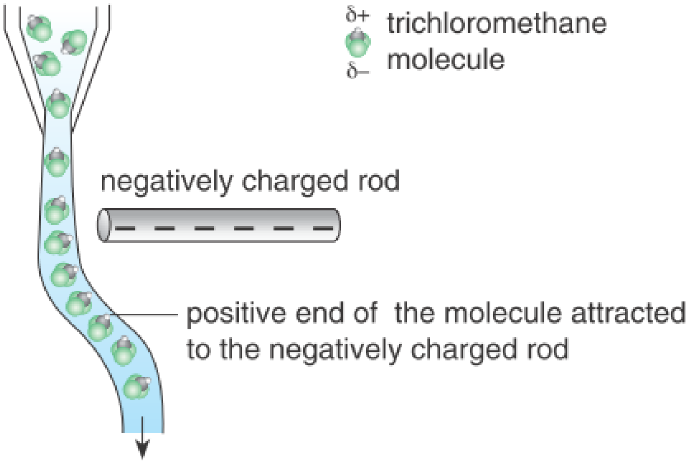
Charged rod experiment

A negatively charged rod is brought near a jet of water running out from a burette. The jet of $CHCl_3$ is deflected as shown:



(a) With reference to the structure of $CHCl_3$, explain why the jet of $CHCl_3$ was deflected.

Answer: Molecular shape of $CHCl_3$ is tetrahedral. Polarities of polar C-H and C-Cl bonds are different. Polarities of polar C-H and C-Cl bonds cannot cancel out. Hence, $CHCl_3$ molecules are polar and are attracted by the electric field.



(b) State the effect on the jet if the negatively charged rod is replaced by a positively charged rod. Explain your answer

Answer: The jet of $CHCl_3$ will be attracted towards the rod. The $CHCl_3$ molecules will orientate themselves so that their negative ends are attracted to the positively charged rod.

(c) If CCl_4 is used instead of $CHCl_3$ and a negatively charged rod is brought near the liquid jet, would the liquid jet be deflected? Explain your answer.

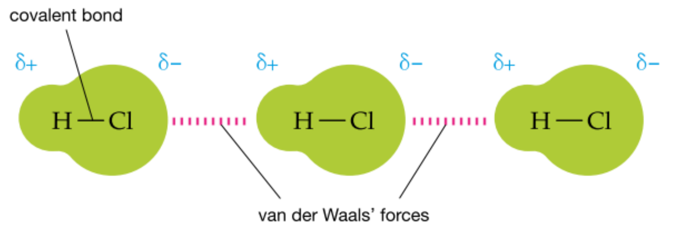
Answer: The jet of CCl_4 does not deflect. Molecular shape of CCl_4 is tetrahedral. Polar C-Cl bonds are arranged symmetrically. Polarities of polar C-Cl bonds can canel out. CCl_4 molecules are non-polar and they are not attracted by the charged rod.

Van der Waals' forces

- An electrostatic attraction that exist between ALL molecules.

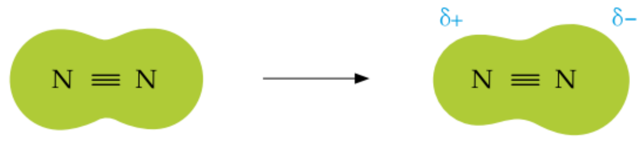
Van der Waals' forces in polar molecules

In a HCl molecule, uneven distribution of shared electrons is permanent as HCl is a polar molecule. The attraction between the positive end of a HCl molecule and the negative end of a HCl molecule belongs to one kind of van der Waals' forces

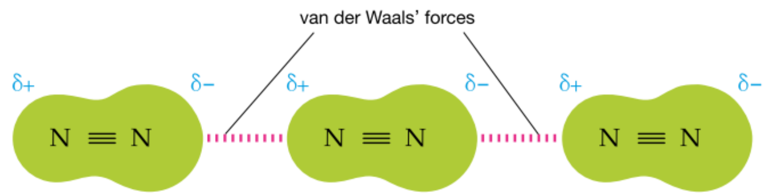


Van der Waals' forces in non-polar molecules

N_2 is a non-polar molecule. Electrons are in constant motion. At a given instant of time, the electron distribution may be uneven. The side with more electrons forms a partial negative charge (δ–) and the side with fewer electrons forms a partial positive charge (δ+). These charges are not permanent. They continue to arise and disappear all the time.



The partial charges on one molecule can affect the electron distribution in neighbouring molecules, inducing a separation of charge on the neighbouring molecules. The attraction between the positive end of a N_2 molecule and the negative end of a N_2 molecule belongs to another kind of van der Waals' forces



Intensive note (Topic 6: Microscopic world II)

Factors affecting strength of van der Waals' forces

1. Molecular size

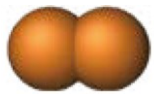
- When there are more electrons in a molecule, there is a higher chance of uneven distribution of electrons in the molecule
- The **larger the molecular size**, the **stronger is the van der Waals' forces** between molecules

F_2

Boiling point = -188°C

Cl_2

Boiling point = -35°C



Br_2

Boiling point = 59°C

I_2

Boiling point = 184°C

2. Molecular shape (usually applicable to carbon compounds)

Pentane [$CH_3CH_2CH_2CH_2CH_3$]

Boiling point = 36.1 °C

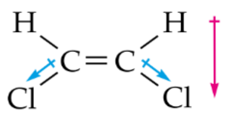
Dimethylpropane [$CH_3C(CH_3)_2CH_3$]

Boiling point = 9.5 °C

- Carbon atoms in pentane molecule are arranged in <u>straight chain</u>. Pentane is a <u>rod-shaped molecule</u>.
- Carbon atoms in dimethylpropane molecule are arranged in a <u>branched chain</u>. Dimethylpropane is a <u>spherical molecule</u>.
- Pentane has a **higher area of contact between molecules**. Van der Waals' forces between pentane molecules are **stronger**.

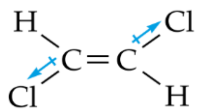
3. Molecular polarity (usually applicable to molecules with comparable sizes)

- Both cis-1,2-dichloroethene and trans-1,2-dichloroethene have 2 polar C-Cl bonds. Since the C–H bonds have a very low polarity, the molecular polarity is mainly determined by the polarity of the C–Cl bonds.



Cis-1,2-dichloroethene

Boiling point = 60.1 °C



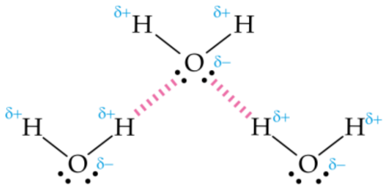
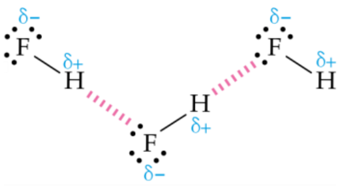
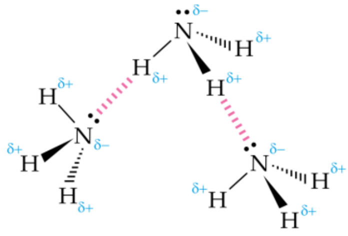
Trans-1,2-dichloroethene

Boiling point = 48.7 °C

- The two polar C–Cl bonds in the cis-1,2-dichloroethene have same polarities and are on the <u>same side of the double bond</u>. As the <u>polarities of these two polar bonds do not cancel out each other</u>, the cis-1,2-dichloroethene molecule is <u>polar</u>.
- The two polar C–Cl bonds in the trans-1,2-dichloroethene have same polarities but are on the <u>opposite sides of the double bond</u>. As the <u>polarities of these two polar bonds cancel out each other</u>, the trans-1,2-dichloroethene molecule is <u>non-polar</u>.
- The van der Waals' forces between **polar** cis-1,2-dichloroethene molecules are **stronger** than those between **non-polar** trans-1,2-dichloroethene molecules.

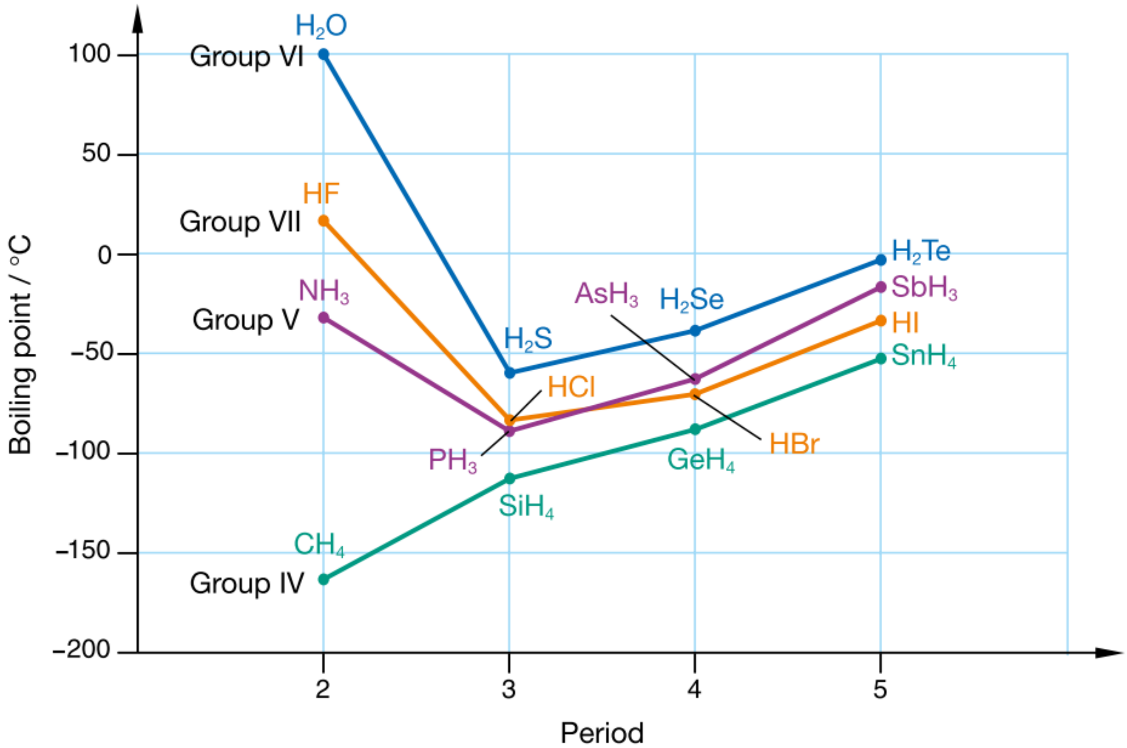
Hydrogen bonds

- When a H atom is bonded to a highly electronegative atom (e.g. N, O, F), a highly polar bond forms.
- Hydrogen bonding forms between the H atom bonded to N / O / F and the lone pair electrons of N / O / F

		
In each H_2O molecule, there are **2 lone pair** electrons on O and **2 O-H** bonds. Each H_2O molecule can form **2 hydrogen bonds** on average	In each HF molecule, there are **3 lone pair** electrons on F and **1 F-H** bonds. Each HF molecule can form **1 hydrogen bonds** on average	In each NH_3 molecule, there are **1 lone pair** electron on N and **3 N-H** bonds. Each NH_3 molecule can form **1 hydrogen bonds** on average

Note: van der Waals' forces also exist between molecule of H_2O, HF and NH_3

Boiling points of hydrides



Explanation on the ascending order of boiling points of HF, HCl and HBr:

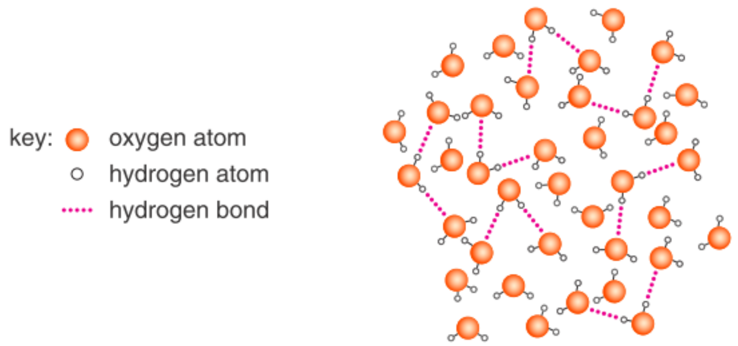
- Both molecules of HCl and HBr are held by weak van der Waals' forces.
- The molecular size of HBr is larger than that of HCl, therefore the van der Waals' forces between HBr molecules are stronger than those between HCl molecules
- Stronger hydrogen bonds exist between HF molecules

Effect of hydrogen bonding on physical properties of water and ethanol

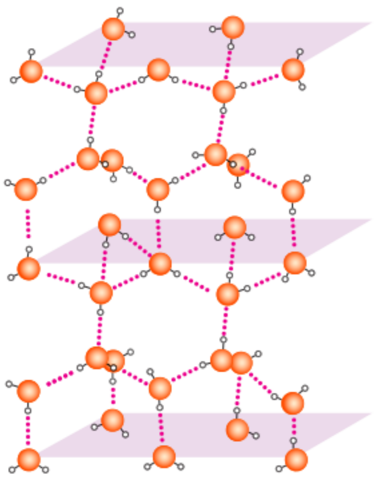
Water	Ethanol
1. High boiling point - Extensive hydrogen bonds between water molecules	1. High boiling point - Extensive hydrogen bonds between ethanol molecules
2. High viscosity [liquid flows slowly] - Extensive hydrogen bonds between water molecules	2. High viscosity [liquid flows slowly] - Extensive hydrogen bonds between ethanol molecules - Less viscous than water
3. High surface tension [Liquid surface to be tightened like an elastic film] - Extensive hydrogen bonds between water molecules	3. Soluble in water - Hydroxyl group (-OH) of ethanol can form hydrogen bonds with water molecules

Density of ice and water

- H_2O molecules are mainly held by strong hydrogen bonds
- In ice, water molecules are arranged tetrahedrally in an open structure because of the extensive hydrogen bonding between them. Thus, they take up a larger volume.
- When ice melts, the water molecules have relative motion and the open structure collapses. The water molecules become more closely packed and they take up a smaller volume.



Arrangement of H_2O molecules in water



Arrangement of H_2O molecules in ice

In ice:

- Molecular shape of H_2O is V-shaped
  (As central O atom still has 2 bond pairs and 2 lone pairs)
- Water molecules are tetrahedrally arranged.
  (2 lone pairs on O atom of each H_2O molecule can form hydrogen bonds with H atoms of 2 neighbouring water molecules, 2 H atoms of each H_2O molecule can form hydrogen bonds with lone pair on O atom of 2 neighbouring water molecules)
- Oxygen atoms in ice are arranged in a hexagonal shape